✪ Psychological Statistics ✪

Analysis of Variance (ANOVA) & ANCOVA

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In [ ]:
```
### [ Setup the working directory ]

setwd("/Users/wesley/[Course]/Python/R_Script")
getwd()
```

In [2]:
```
### [ Loading the required libraries ]

library("dplyr")
library("rstatix")
library("ggpubr")
library("pastecs")
library("compute.es")
library("emmeans")
library("multcomp")
library("agricolae")
```

---

(1) One-way ANOVA

Example A: Libido & Dosage of Viagra

[Hypothesis] Dosage of viagra affects one's libido level. *(2-tailed)*

- Null hypothesis H_0 **:** There are no difference among the 3 independent groups (1 factor: dosage).
- Alternative hyp. H_1 **:** There are difference among the 3 independent groups (Placebo, Low dose, High dose).

In [5]:
```
### [ Step.1 ] Load data

libido <- c(3,2,1,1,4,5,2,4,2,3,7,4,5,3,6)
    #dose <- c(rep(1,5),rep(2,5), rep(3,5))
    #dose <- factor(dose, levels = c(1:3), labels = c("Placebo", "Low Dose", "H
dose <- gl(3,5, labels = c("Placebo", "Low Dose", "High Dose"))
viagraData <- data.frame(dose, libido)
```

In [6]:
```
### [ Step.2 ] Check assumptions

#----- (a) Outliers -----#
viagraData %>%
    group_by(dose) %>%
    identify_outliers(libido)

#----- (b) Normality -----#
viagraData %>%
    group_by(dose) %>%
    shapiro_test(libido)

#----- (c) Homogeneity of Variance -----#
viagraData %>%
    levene_test(libido ~ dose)
```

A data.frame: 0 × 4

dose	libido	is.outlier	is.extreme
<fct>	<dbl>	<lgl>	<lgl>

A tibble: 3 × 4

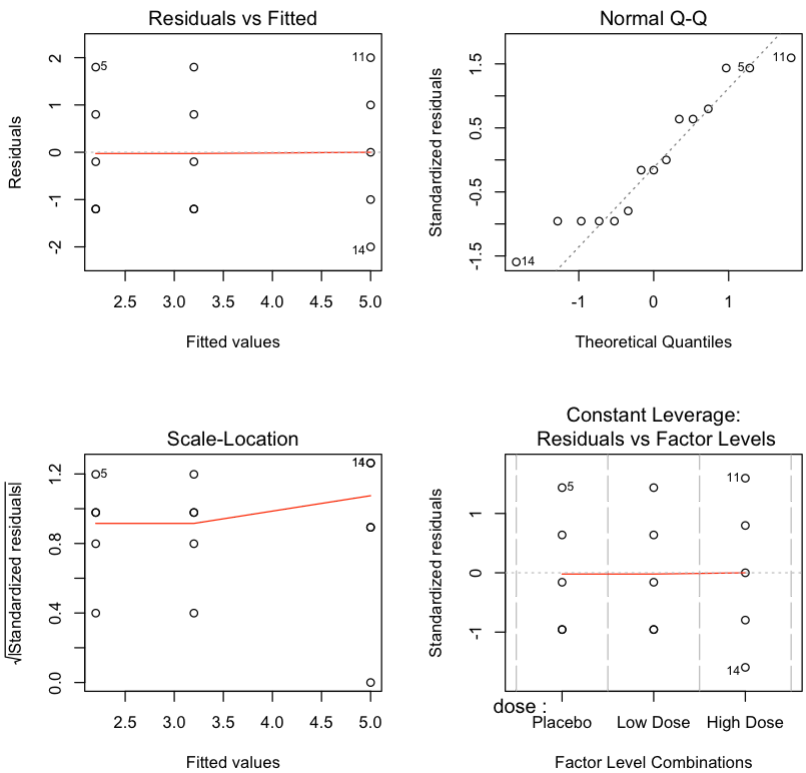
dose	variable	statistic	p
<fct>	<chr>	<dbl>	<dbl>
Placebo	libido	0.9020198	0.4211497
Low Dose	libido	0.9020198	0.4211497
High Dose	libido	0.9867622	0.9671739

A tibble: 1 × 4

df1	df2	statistic	p
<int>	<int>	<dbl>	<dbl>
2	12	0.1176471	0.8900225

In [7]:
```
### [ Step.3-1 ] ANOVA

viagraModel <- aov(libido~dose, data = viagraData)
#viagraModel_lm<-lm(libido~dose, data = viagraData)

summary(viagraModel)
summary.lm(viagraModel)

par(mfrow=c(2,2))
plot(viagraModel)
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
dose         2  20.13  10.067   5.119 0.0247 *
Residuals   12  23.60   1.967
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Call:
aov(formula = libido ~ dose, data = viagraData)

Residuals:
   Min     1Q Median     3Q    Max
  -2.0   -1.2   -0.2    0.9    2.0

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)     2.2000     0.6272   3.508  0.00432 **
doseLow Dose    1.0000     0.8869   1.127  0.28158
doseHigh Dose   2.8000     0.8869   3.157  0.00827 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.402 on 12 degrees of freedom
Multiple R-squared:  0.4604,    Adjusted R-squared:  0.3704
F-statistic: 5.119 on 2 and 12 DF,  p-value: 0.02469
```



In [26]:
```
### [ Step.3-1 ] Welch's ANOVA (When variances are not equal across groups)

oneway.test(libido ~ dose, data = viagraData)
```

```
        One-way analysis of means (not assuming equal variances)

data:  libido and dose
F = 4.3205, num df = 2.0000, denom df = 7.9434, p-value = 0.05374
```

In [57]:
```
### [ Step.3-2 ] Post-hoc Tests

#--- (a) Planned Contrasts ---#

contrasts(viagraData$dose)<-cbind(c(-2,1,1), c(0, -1, 1))
viagraPlanned<-aov(libido~dose, data = viagraData)
summary.lm(viagraPlanned)
summary(viagraPlanned)

# contrasts(viagraData$dose)<-contr.sum(3)
# viagraModel2<-aov(libido~dose, data = viagraData)
# summary.lm(viagraModel2)

# contrasts(viagraData$dose)<-contr.helmert(3)
# viagraModel2<-aov(libido~dose, data = viagraData)
# summary.lm(viagraModel2)
```

```
Call:
aov(formula = libido ~ dose, data = viagraData)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.8462 -1.2222 -0.8462  1.1538  4.1538 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  4.31446    0.34821  12.390 1.19e-12 ***
dose1        0.54612    0.25018   2.183   0.0379 *  
dose2       -0.01442    0.41950  -0.034   0.9728    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.867 on 27 degrees of freedom
Multiple R-squared:  0.1518,	Adjusted R-squared:  0.08896 
F-statistic: 2.416 on 2 and 27 DF,  p-value: 0.1083

            Df Sum Sq Mean Sq F value Pr(>F)
dose         2  16.84   8.422   2.416  0.108
Residuals   27  94.12   3.486
```

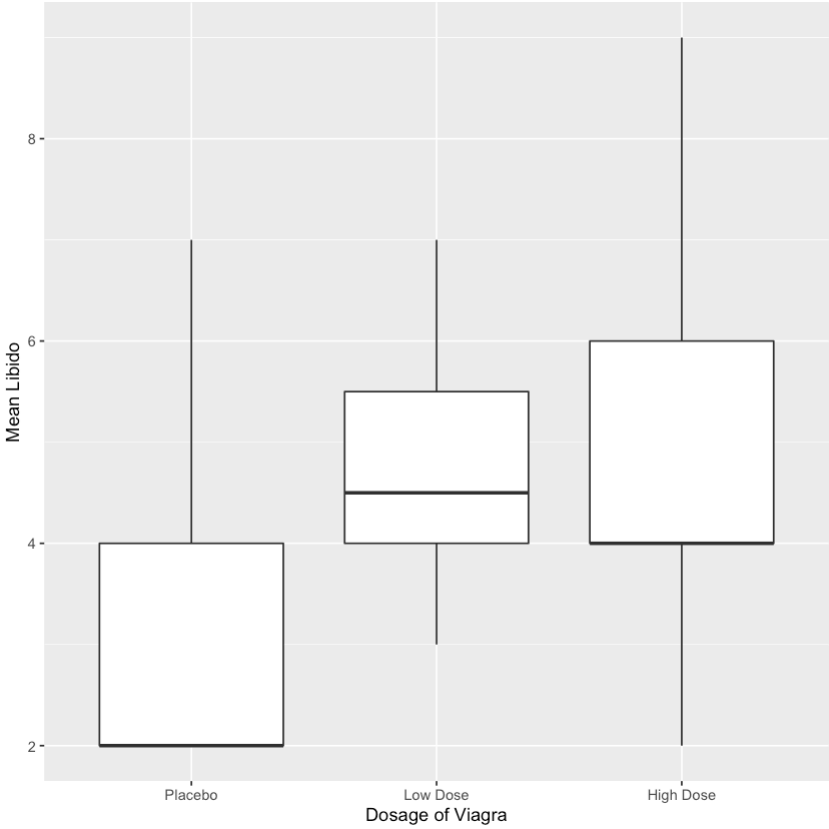
In [71]:
```
### [ Step.3-2 ] Post-hoc Tests

#--- (b) Pairwise comparisons ---#

pairwise.t.test(viagraData$libido, viagraData$dose, p.adjust.method = "bonferro
# pairwise.t.test(viagraData$libido, viagraData$dose, p.adjust.method = "fdr")

#TukeyHSD(viagraModel)

#library(multcomp)
#postHocs<-glht(viagraModel, linfct = mcp(dose = "Tukey"))
#summary(postHocs)
#confint(postHocs)

#library(agricolae)
#vigcomp <- scheffe.test(viagraModel,"dose", group=TRUE,console=TRUE)
#print(vigcomp$comparison)
#plot(vigcomp)
```

```
        Pairwise comparisons using t tests with pooled SD

data:  viagraData$libido and viagraData$dose

          Placebo Low Dose
Low Dose  0.24    -
High Dose 0.16    1.00

P value adjustment method: bonferroni
```

In [35]:
```
### [ Step.4 ] Effect size

omega<-function(SSm, SSr, dfm, MSr)
{
    SSt = SSm + SSr
    omega = (SSm-(dfm*MSr))/(SSt+MSr)
    print(paste("Omega-Squared: ", omega))
}

omega(20.133, 23.600, 2, 1.9667)
omega(450.66,38.09, 5, 0.334)
omega(4180.6, 4356.6, 3, 167.56)

#Effect size from post hoc mean values of each group
#compute.es::mes(2.2, 3.2, 1.3038405, 1.3038405, 5, 5)
#compute.es::mes(2.2, 5, 1.3038405, 1.5811388, 5, 5)
#compute.es::mes(3.2, 5, 1.3038405, 1.5811388, 5, 5)

#Effect size from post hoc t values of each group
#compute.es::tes(1.127, 5, 5)
#compute.es::tes(3.157, 5, 5)
#compute.es::tes(2.029, 5, 5)
```

```
[1] "Omega-Squared:  0.35447935106795"
[1] "Omega-Squared:  0.918022262024519"
[1] "Omega-Squared:  0.422518254380362"
```

In [82]:
```
### [ Step.5 ] Visualization

ggplot(viagraData, aes(x = dose, y = libido)) + geom_boxplot() +
  labs(x = "Dosage of Viagra", y = "Mean Libido")
```



~ *Report* ~

- There was a significant effect of Viagra on levels of libido, $F_{2,12} = 5.12$, $p < .05$, $\omega^2 = .35$.
- Bonferroni tests revealed non-significant differences between low-dose group and both the placebo ($p = .845$, $d = -0.77$) and high-dose ($p = .196$, $d = -1.24$) groups.
- The high-dose group, however, had a mean almost 2 SD bigger than the placebo group ($p = .025$, $d = -1.93$).

Example B

[Hypothesis] Specific treatments (control, tr1, tr2) can enhance plant growth. *(1-tailed)*

- Null hypothesis H_0 **:** No prominent treatment effect on plant growth.
- Alternative hyp. H_1 **:** The treatment can enhance plant growth.

In [41]:
```
### [ Step.1 ] Data Loading

# → Loading the dataset
data("PlantGrowth")
set.seed(1234)
PlantGrowth %>% head(3)

levels(PlantGrowth$group)

# If the levels are not automatically in the correct order, re-order them as fo
PlantGrowth <- PlantGrowth %>%
  reorder_levels(group, order = c("ctrl", "trt1", "trt2"))

# Descriptive statistics
PlantGrowth %>%
  group_by(group) %>%
  get_summary_stats(weight, type = "mean_sd")
```

A data.frame: 3 × 2

	weight	group
	<dbl>	<fct>
1	4.17	ctrl
2	5.58	ctrl
3	5.18	ctrl

'ctrl' · 'trt1' · 'trt2'

A tibble: 3 × 5

group	variable	n	mean	sd
<fct>	<chr>	<dbl>	<dbl>	<dbl>
ctrl	weight	10	5.032	0.583
trt1	weight	10	4.661	0.794
trt2	weight	10	5.526	0.443

In [42]:
```
### [ Step.2 ] Check assumptions

#-----(a) Outliers -----#
PlantGrowth %>%
  group_by(group) %>%
  identify_outliers(weight)
```

A tibble: 2 × 4

group	weight	is.outlier	is.extreme
<fct>	<dbl>	<lgl>	<lgl>
trt1	5.87	TRUE	FALSE
trt1	6.03	TRUE	FALSE

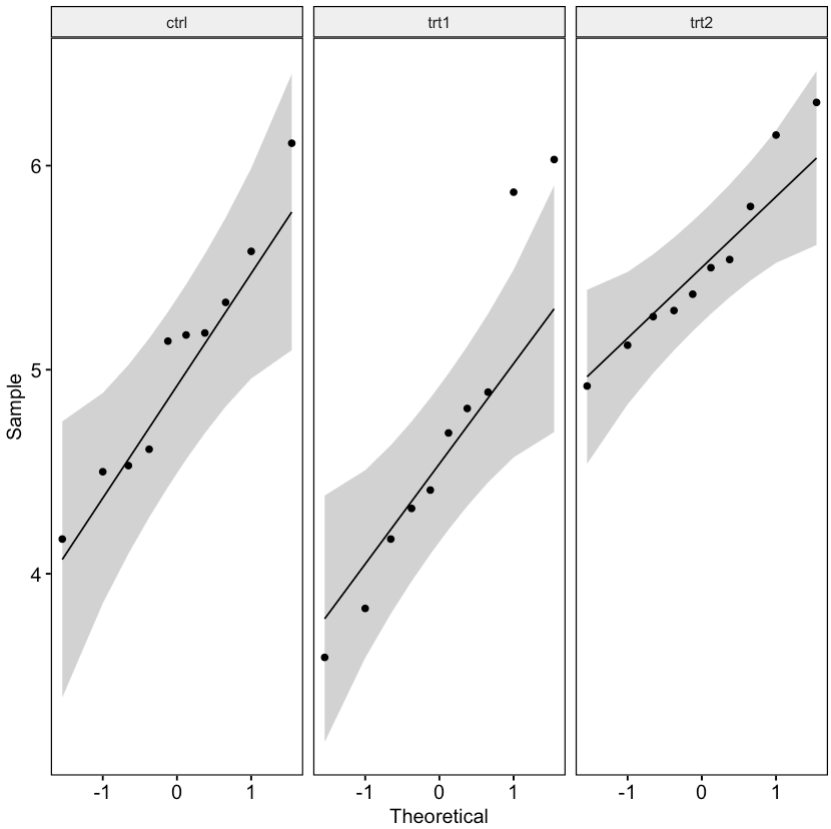
In [43]:
```
### [ Step.2 ] Check assumptions

#-----(b) Normality assumption -----#

# Compute Shapiro-Wilk test of normality
shapiro_test(residuals(model))

PlantGrowth %>%
  group_by(group) %>%
  shapiro_test(weight)

# visual inspection
model  <- lm(weight ~ group, data = PlantGrowth)
# Create a QQ plot of residuals
#ggqqplot(residuals(model))
# All points in QQ plot fall along the reference line, we can assume normality.
ggqqplot(PlantGrowth, "weight", facet.by = "group")
```

A tibble: 1 × 3

variable	statistic	p.value
<chr>	<dbl>	<dbl>
residuals(model)	0.9753224	0.4440646

A tibble: 3 × 4

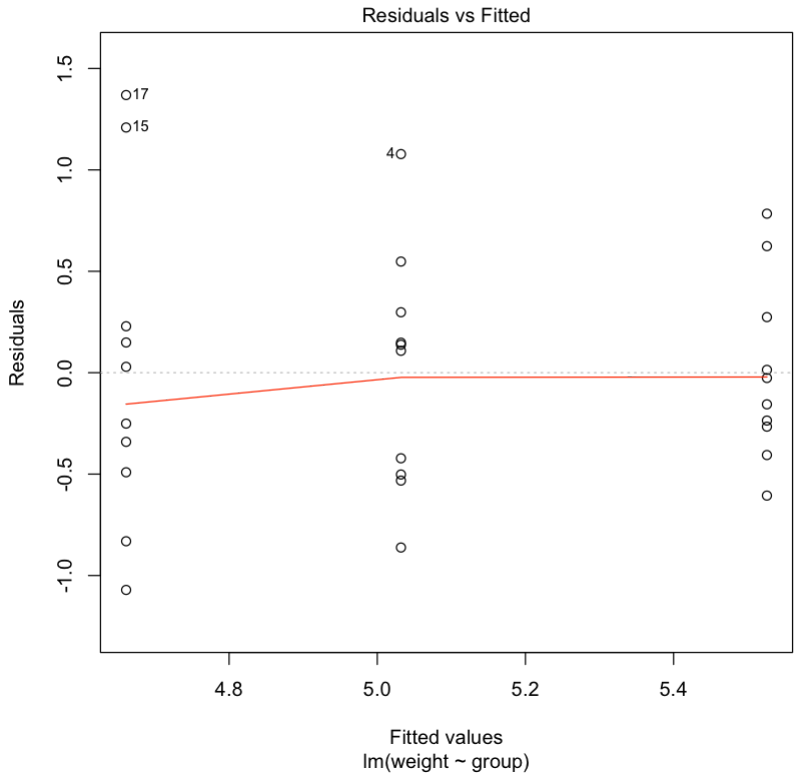
group	variable	statistic	p
<fct>	<chr>	<dbl>	<dbl>
ctrl	weight	0.9566815	0.7474734
trt1	weight	0.9304107	0.4519440
trt2	weight	0.9410052	0.5642519



In [44]:
```
### [ Step.2 ] Check assumptions

#-----(c) Homogeneity of Variance -----#

plot(model, 1)

PlantGrowth %>% levene_test(weight ~ group)
```

A tibble: 1 × 4

df1	df2	statistic	p
<int>	<int>	<dbl>	<dbl>
2	27	1.119186	0.3412266



In [45]:
```
### [ Step.3-1 ] ANOVA

(res.aov <- PlantGrowth %>% anova_test(weight ~ group))

(res.aov2 <- PlantGrowth %>% welch_anova_test(weight ~ group))
```

```
Coefficient covariances computed by hccm()
```

A anova_test: 1 × 7

	Effect	DFn	DFd	F	p	p<.05	ges
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	group	2	27	4.846	0.016	*	0.264

A rstatix_test: 1 × 7

	.y.	n	statistic	DFn	DFd	p	method
	<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	weight	30	5.18	2	17.12842	0.017	Welch ANOVA

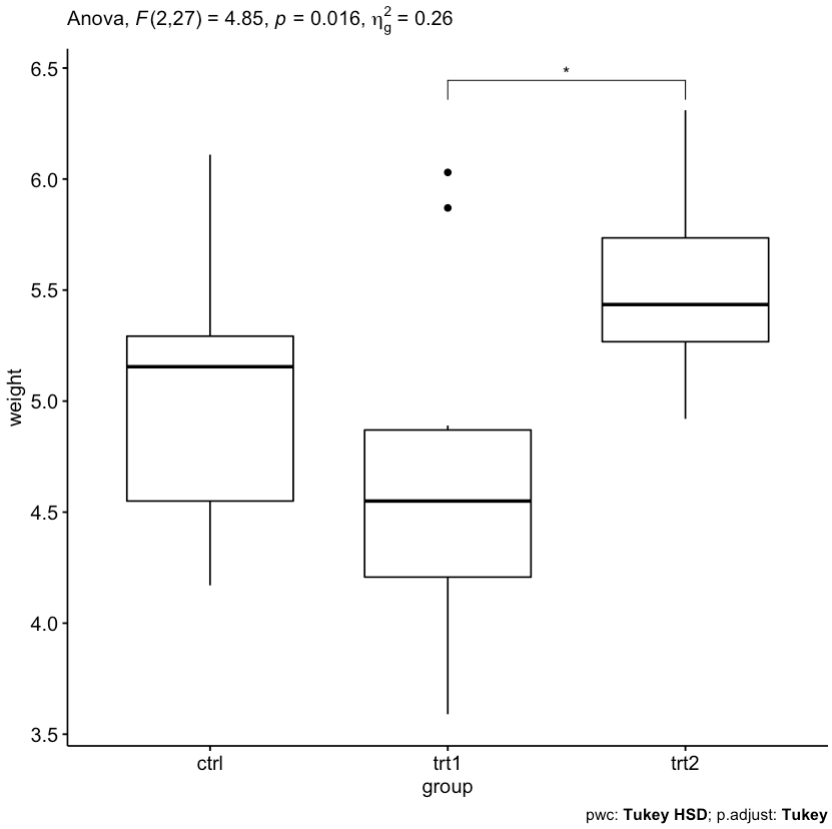
In [46]:
```
### [ Step.3-2 ] post-hoc test

(pwc <- PlantGrowth %>% tukey_hsd(weight ~ group))
```

A rstatix_test: 3 × 9

	term	group1	group2	null.value	estimate	conf.low	conf.high	p.adj	p.adj.signif
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	group	ctrl	trt1	0	-0.371	-1.0622161	0.3202161	0.391	ns
2	group	ctrl	trt2	0	0.494	-0.1972161	1.1852161	0.198	ns
3	group	trt1	trt2	0	0.865	0.1737839	1.5562161	0.012	*

In [ ]:
```
### [ Step.4 ] Effect size

# ----- This has been shown in the ANOVA table -----#
```

In [47]:
```
### [ Step.5 ] Visualization

pwc <- pwc %>% add_xy_position(x = "group")

ggboxplot(PlantGrowth, x = "group", y = "weight") +
  stat_pvalue_manual(pwc, hide.ns = TRUE) +
  labs(subtitle = get_test_label(res.aov, detailed = TRUE),
    caption = get_pwc_label(pwc))
```



~ *Report* ~

- Plant growth was statistically significantly different between different treatment groups, $F_{2, 27} = 4.85$, $p = 0.016$, *generalized* $\eta^2 = 0.26$.
- Plant growth decreased in trt1 group (4.66 +/- 0.79) compared to ctr group (5.03 +/- 0.58). It increased in trt2 group (5.53 +/- 0.44) compared to trt1 and ctr group.
- Tukey post-hoc analyses revealed that the increase from trt1 to trt2 (0.87, 95% CI (0.17 to 1.56)) was statistically significant ($p = 0.012$), but no other group differences were statistically significant.

---

(2) ANCOVA

Example A: Libido and Dosage of Viagra

[Hypothesis] One's libido level is affected by the dosage of viagra and the partner's libido. *(2-tailed)*

- Null hypothesis H_0 **:** Controlling partner's libido, there are no difference among the 3 independent groups (1 factor: dosage).
- Alternative hyp. H_1 **:** Controlling partner's libido, there are difference among the 3 independent groups (Placebo, Low dose, High dose).

In [9]:
```
### [ Step.1 ] Data Loading

# → Loading the dataset
libido<-c(3,2,5,2,2,2,7,2,4,7,5,3,4,4,7,5,4,9,2,6,3,4,4,4,6,4,6,2,8,5)
partnerLibido<-c(4,1,5,1,2,2,7,4,5,5,3,1,2,2,6,4,2,1,3,5,4,3,3,2,0,1,3,0,1,0)
dose<-c(rep(1,9),rep(2,8), rep(3,13))
dose<-factor(dose, levels = c(1:3), labels = c("Placebo", "Low Dose", "High Dos
viagraData<-data.frame(dose, libido, partnerLibido)
```

In [10]:
```
### [ Step.2 ] Check assumptions

#----- (a) Outliers -----#
viagraData %>% group_by(dose) %>%
    identify_outliers(libido)

#----- (b) Normality -----#
viagraData %>% group_by(dose) %>%
    shapiro_test(libido)

#----- (c) Homogeneity of Variance -----#
viagraData %>% levene_test(libido ~ dose)

#----- (d) Check whether the IV and covariate are independent -----#
checkIndependenceModel<-aov(partnerLibido ~ dose, data = viagraData)
summary(checkIndependenceModel)
```

A data.frame: 0 × 5

dose	libido	partnerLibido	is.outlier	is.extreme
<fct>	<dbl>	<dbl>	<lgl>	<lgl>

A tibble: 3 × 4

dose	variable	statistic	p
<fct>	<chr>	<dbl>	<dbl>
Placebo	libido	0.7601871	0.007114315
Low Dose	libido	0.8716216	0.156325133
High Dose	libido	0.9349416	0.394810520

A tibble: 1 × 4

df1	df2	statistic	p
<int>	<int>	<dbl>	<dbl>
2	27	0.3255637	0.7249156

```
            Df Sum Sq Mean Sq F value Pr(>F)
dose         2  12.77   6.385   1.979  0.158
Residuals   27  87.10   3.226
```

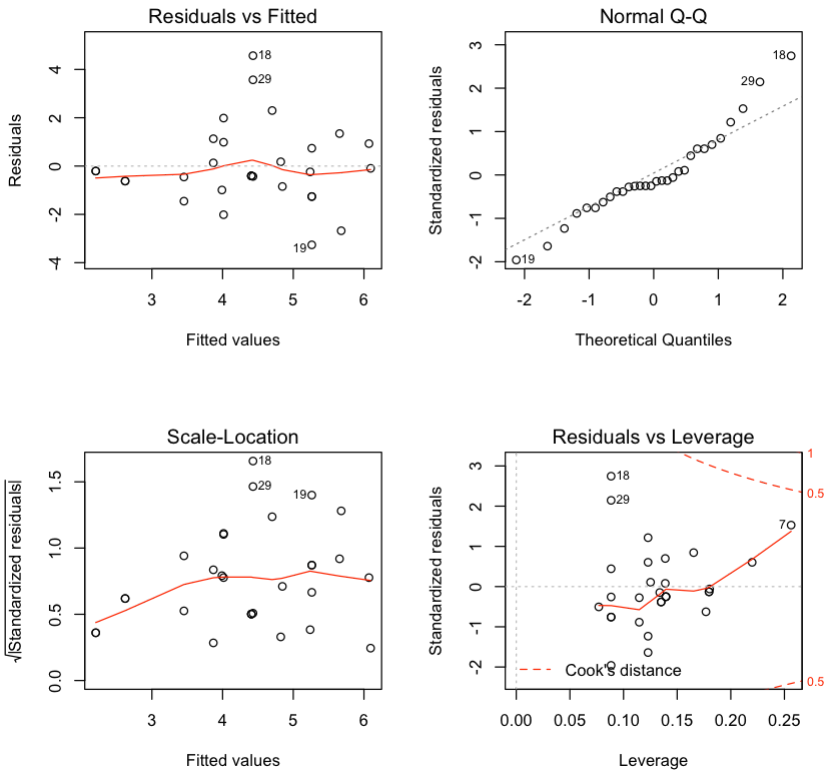
In [11]:
```
### [ Step.3-1 ] ANCOVA

contrasts(viagraData$dose)<-cbind(c(-2,1,1), c(0,-1,1))
viagraModel<-aov(libido~ partnerLibido + dose, data = viagraData)
#summary(viagraModel)
Anova(viagraModel)
```

A anova: 3 × 4

	Sum Sq	Df	F value	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>
partnerLibido	15.07575	1	4.958681	0.03483338
dose	25.18519	2	4.141929	0.02744654
Residuals	79.04712	26	NA	NA

In [12]:
```
### [ Step.3-2 ] Post-hoc tests

TukeyHSD(viagraModel, "dose")

# postHocs<-glht(viagraModel, linfct = mcp(dose = "Tukey"))
# summary(postHocs)
# confint(postHocs)

par(mfrow=c(2,2))
plot(viagraModel)
```

```
Warning message in replications(paste("~", xx), data = mf):
“non-factors ignored: partnerLibido”

  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = libido ~ partnerLibido + dose, data = viagraData)

$dose
                        diff        lwr      upr     p adj
Low Dose-Placebo   1.7357310 -0.3696097 3.841072 0.1207019
High Dose-Placebo  1.9990243  0.1202155 3.877833 0.0353846
High Dose-Low Dose 0.2632934 -1.6836672 2.210254 0.9397954
```



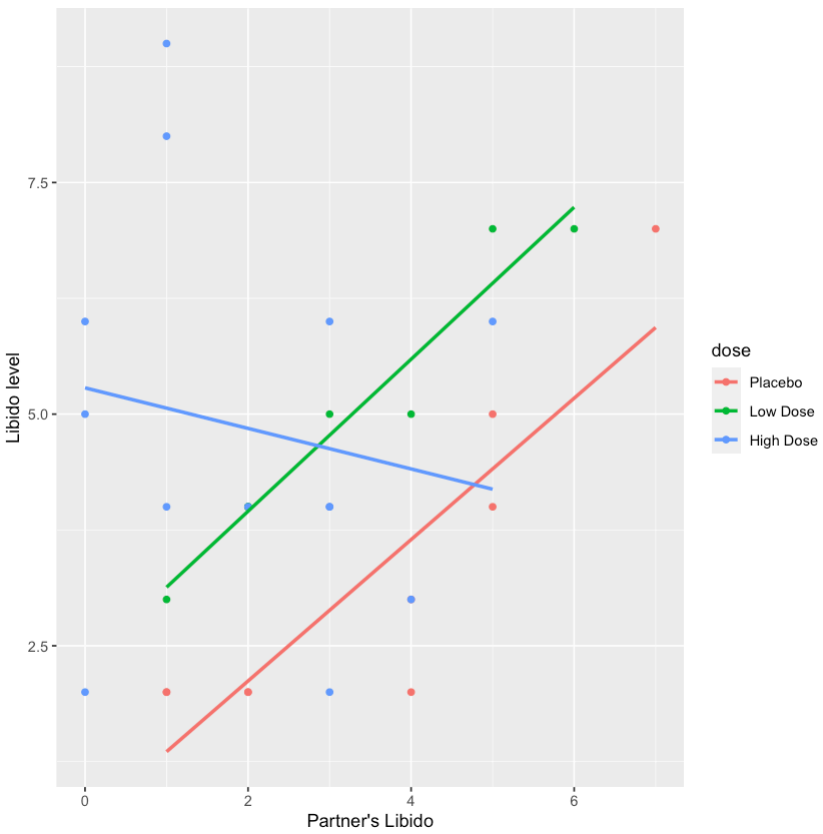
In [13]:
```
### [ Step.2 ] Check assumptions

#-----(e) Homogeneity of regression slopes -----#
# this indicates that partnerLibido shall not have "interaction" among differen

#hoRS<-aov(libido ~ dose + partnerLibido + partnerLibido:dose, data = viagraDat
#hoRS<-aov(libido ~ partnerLibido*dose, data = viagraData)
hoRS<-update(viagraModel, .~. + partnerLibido:dose)
Anova(hoRS, type="III")
```

A anova: 5 × 4

	Sum Sq	Df	F value	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	53.54187	1	21.920735	0.0000932259
partnerLibido	17.18222	1	7.034625	0.0139474621
dose	36.55756	2	7.483569	0.0029795645
partnerLibido:dose	20.42659	2	4.181456	0.0276671129
Residuals	58.62052	24	NA	NA

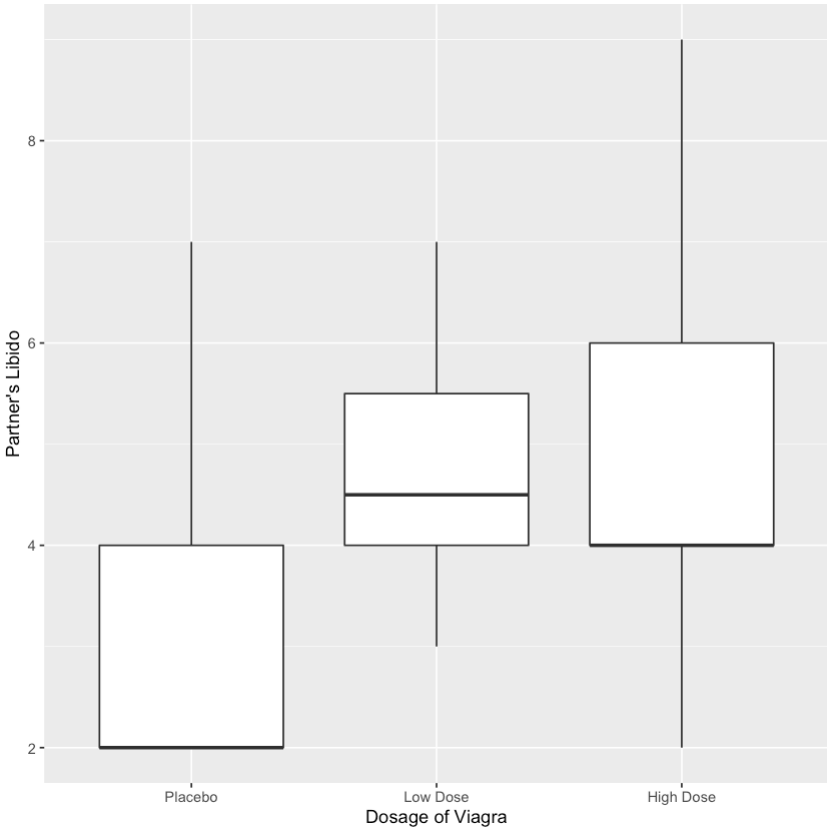
In [19]:
```
ggplot(viagraData, aes(x = partnerLibido, y = libido)) +
    geom_point(aes(color = dose)) +
    geom_smooth(method=lm, se = FALSE, aes(color = dose))+
    labs(x = "Partner's Libido", y = "Libido level")
```

```
`geom_smooth()` using formula 'y ~ x'
```



In [55]:
```
### [ Step.4 ] Effect size

#Effect size from post hoc tests: mean values from Adjusted means
#mes(5.988117, 4.151886, 1.755879, 1.788613, 8, 9)
#mes(6.184427, 4.151886, 1.812267, 1.788613, 13, 9)
#mes(6.184427, 5.988117, 1.812267, 1.755879, 13, 8)

#Effect size from post hoc t values
#tes(2.102, 9, 8)
#tes(2.771, 9, 13)
#tes(0.541, 8, 13)
```

In [21]:
```
### [ Step.5 ] Visualization

ggplot(viagraData, aes(x = dose, y = libido)) +
  geom_boxplot() +
  labs(x = "Dosage of Viagra", y = "Partner's Libido")
```



~ *Report* ~

- The covariate, partner's libido, was not significantly related to the viagra dosage, $F_{2, 26} = 1.98$, $p > .58$.
- There was a significant effect of the dose of Viagra on levels of libido after controlling for the effect of partner's libido, $F_{2, 26} = 4.14$, $p < .05$, *partial* $\eta^2 = .24$.
- Planned contrasts revealed that taking a high or low dose of Viagra significantly increased libido compared to taking a placebo, $t_{26} = 2.79$, $p < .01$, $r = .48$; there was no significant difference between the high and low doses of Viagra, $t_{26} = 0.54$, $p = .59$, $r = .11$.

Example B:

[Hypothesis] Effect of exercises (grp1: low, grp2: moderate and grp3: high) reduces the level of anxiety, which may depend on their pre-test measure. *(1-tailed)*

- Null hypothesis H_0 **:** Controlling the pre-test measure, no prominent difference can be observed among the 3 exercise groups.
- Alternative hyp. H_1 **:** Controlling the pre-test measure, there is prominent difference among the 3 exercise groups.

In [36]:
```
### [ Step.1 ] Data loading

data("anxiety", package = "datarium")

anxiety <- anxiety %>% dplyr::select(id, group, t1, t3) %>%
            rename(pretest = t1, posttest = t3)

anxiety[14, "posttest"] <- 19
```

In [37]:
```
# Inspect the data by showing one random row by groups
set.seed(123)
#anxiety %>% sample_n_by(group, size = 1)

#----- Descriptive stat. -----
anxiety %>%
  group_by(group) %>%
  get_summary_stats(pretest, posttest, type = "mean_sd")
```

A tibble: 6 × 5

group	variable	n	mean	sd
<fct>	<chr>	<dbl>	<dbl>	<dbl>
grp1	posttest	15	16.620	1.676
grp1	pretest	15	17.087	1.629
grp2	posttest	15	15.527	1.703
grp2	pretest	15	16.647	1.566
grp3	posttest	15	13.560	1.423
grp3	pretest	15	17.013	1.321

In [38]:
```
### [ Step.2 ] Check assumptions

#----- Linearity assumption -----#
#ggscatter(anxiety, x = "pretest", y = "posttest", color = "group", add = "reg.
#stat_regline_equation(aes(label = paste(..eq.label.., ..rr.label.., sep = "~~~
#                           color = group))

#----- (a) Normality -----#
# Build the linear model, the covariate goes first
model <- lm(posttest ~ pretest + group, data = anxiety)

# Inspect the model diagnostic metrics
model.metrics <- augment(model) %>%
  dplyr::select(-.hat, -.sigma, -.fitted, -.sigma) # Remove details
head(model.metrics, 3)

# Compute Shapiro-Wilk test of normality
shapiro_test(residuals(model))
# shapiro_test(model.metrics$.resid)

#----- (b) Outliers -----#
model.metrics %>%
  dplyr::filter(abs(.std.resid) > 3) %>%
  as.data.frame()

#----- (c) Homogeneity of Variance -----#
plot(model, 1)
model.metrics %>% levene_test(.resid ~ group)

#----- (d) Independence of Covariate -----#
checkIndep <- aov(pretest ~ group, data = model.metrics)
summary(checkIndep)

#----- (e) Homogeneity of Regression Slope -----#
anxiety %>% anova_test(posttest ~ group*pretest)
```

A tibble: 3 × 6

posttest	pretest	group	.resid	.cooksd	.std.resid
<dbl>	<dbl>	<fct>	<dbl>	<dbl>	<dbl>
14.1	14.1	grp1	0.5495570	0.10086429	1.4570682
14.3	14.5	grp1	0.3384556	0.03095139	0.8852028
14.9	15.7	grp1	-0.2948486	0.01334801	-0.7498662

A tibble: 1 × 3

variable	statistic	p.value
<chr>	<dbl>	<dbl>
residuals(model)	0.9753224	0.4440646

A data.frame: 0 × 6

posttest	pretest	group	.resid	.cooksd	.std.resid
<dbl>	<dbl>	<fct>	<dbl>	<dbl>	<dbl>

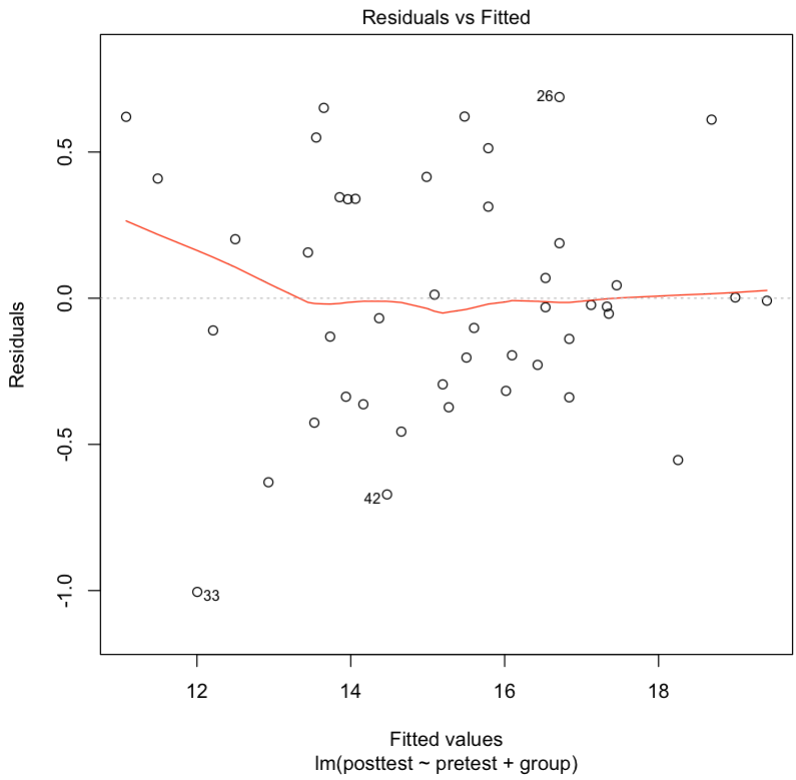
A tibble: 1 × 4

df1	df2	statistic	p
<int>	<int>	<dbl>	<dbl>
2	42	2.271753	0.1156629

```
            Df Sum Sq Mean Sq F value Pr(>F)
group        2   1.67  0.8336   0.365  0.696
Residuals   42  95.89  2.2831

Coefficient covariances computed by hccm()
```

A anova_test: 3 × 7

	Effect	DFn	DFd	F	p	p<.05	ges
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	group	2	39	209.314	1.40e-21	*	0.915
2	pretest	1	39	572.828	6.36e-25	*	0.936
3	group:pretest	2	39	0.127	8.81e-01		0.006



In [39]:
```
### [ Step.3-1 ] ANCOVA

res.aov <- anxiety %>% anova_test(posttest ~ pretest + group)
#res.aov
get_anova_table(res.aov)
```

```
Coefficient covariances computed by hccm()
```

A anova_test: 2 × 7

	Effect	DFn	DFd	F	p	p<.05	ges
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	pretest	1	41	598.321	4.48e-26	*	0.936
2	group	2	41	218.629	1.35e-22	*	0.914

In [40]:
```
### [ Step.3-2 ] Post-hoc test

#-----> Pairwise comparisons
pwc <- anxiety %>%
  emmeans_test(posttest ~ group, covariate = pretest, p.adjust.method = "bonfe
pwc

# Display the adjusted means of each group
# Also called as the estimated marginal means (emmeans)
get_emmeans(pwc)
```

A rstatix_test: 3 × 9

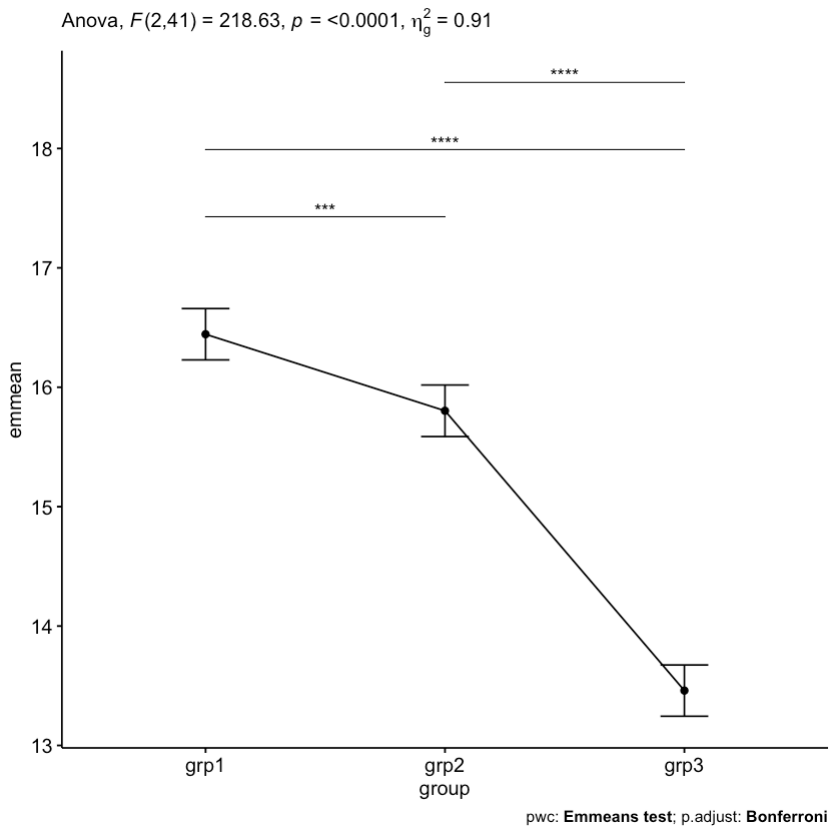
	term	.y.	group1	group2	df	statistic	p	p.adj	p.adj.signif
	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	pretest*group	posttest	grp1	grp2	41	4.235403	1.257090e-04	3.771269e-04	***
2	pretest*group	posttest	grp1	grp3	41	19.861727	1.192286e-22	3.576859e-22	****
3	pretest*group	posttest	grp2	grp3	41	15.517184	9.207816e-19	2.762345e-18	****

A tibble: 3 × 8

pretest	group	emmean	se	df	conf.low	conf.high	method
<dbl>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
16.91556	grp1	16.44414	0.1064779	41	16.22910	16.65918	Emmeans test
16.91556	grp2	15.80302	0.1068340	41	15.58726	16.01877	Emmeans test
16.91556	grp3	13.45951	0.1063143	41	13.24480	13.67421	Emmeans test

In [21]:
```
### [ Step.4 ] Effect size -- Generalized Eta Squared

# ----- This has been shown in the ANOVA table -----#
```

In [22]:
```
### [ Step.5 ] Visualization

pwc <- pwc %>% add_xy_position(x = "group", fun = "mean_se")
ggline(get_emmeans(pwc), x = "group", y = "emmean") +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0.2) +
  stat_pvalue_manual(pwc, hide.ns = TRUE, tip.length = FALSE) +
  labs(subtitle = get_test_label(res.aov, detailed = TRUE),
       caption = get_pwc_label(pwc))
```



~ *Report* ~

- After adjustment for pre-test anxiety score, there was a statistically significant difference in post-test anxiety score between the groups, $F_{2,\ 41} = 218.63$, $p < 0.001$.
- Post hoc analysis was performed with a Bonferroni adjustment. The mean anxiety score was statistically significantly greater in grp1 (16.4 +/- 0.15) compared to the grp2 (15.8 +/- 0.12) and grp3 (13.5 +/_ 0.11), $p < 0.001$.